

SMART PARKING SYSTEM

Project Report

Arduino-Based Parking Management System

This project uses IR sensors, Arduino Uno, LCD display, LEDs and a servo motor. Camera and IoT are not required.

Submitted as a Project / Seminar Report

1. Introduction

A **Smart Parking System** is an automated system designed to manage parking spaces efficiently. It detects whether parking slots are free or occupied and provides information to the user.

In a traditional parking system, drivers have to search manually for an empty parking space. This takes time and may cause unnecessary traffic and fuel consumption.

The proposed system uses **IR sensors, Arduino, LCD display, LEDs, and a servo motor**. The sensors detect vehicles in each parking slot. Arduino processes the sensor information and displays the number of available slots.

The system is simple, economical, and suitable for small parking areas such as colleges, offices, hospitals, and shopping centers.

2. Problem Statement and Objectives

Problem Statement

Parking has become a common problem in crowded areas. Drivers often spend considerable time searching for an empty parking space.

Problems in traditional parking:

- Difficulty finding empty slots
- Wastage of time
- Unnecessary fuel consumption
- Traffic congestion
- Manual monitoring
- Poor utilization of parking spaces

Proposed Solution

The Smart Parking System automatically detects the availability of parking slots using sensors. The information is processed by an Arduino and displayed on an LCD.

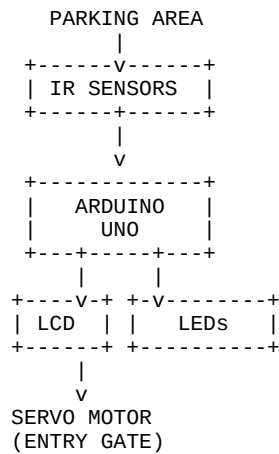
Objectives

1. Detect available parking slots.
2. Detect occupied slots.
3. Display the number of available spaces.
4. Reduce parking search time.
5. Reduce unnecessary vehicle movement.
6. Reduce manual parking management.

3. Components Required

- 1. Arduino Uno:** The main controller. It receives sensor signals and controls the LCD, LEDs, and servo motor.
- 2. IR Sensors:** Detect whether a vehicle is present in a parking slot.
- 3. LCD Display:** A 16×2 LCD displays information such as the number of available parking slots.
- 4. Servo Motor:** Used to open and close the parking entrance gate.
- 5. LEDs:** Indicate the status of parking slots. Green can indicate free and red can indicate occupied.
- 6. Jumper Wires:** Used for electrical connections.
- 7. Breadboard:** Used to assemble the circuit without soldering.
- 8. Power Supply:** Provides electrical power to the Arduino and other components.

4. Block Diagram



Block Description

IR Sensors: Detect vehicles in parking slots.

Arduino: Processes sensor signals and controls the system.

LCD: Displays the number of available slots.

LEDs: Indicate individual slot status.

Servo Motor: Controls the parking entrance gate.

5. Working Principle

The Smart Parking System works by detecting vehicles using IR sensors.

When the system is switched ON, Arduino checks all the parking-slot sensors. If no vehicle is detected, the parking slot is considered **FREE**. If a vehicle is detected, the slot is considered **OCCUPIED**.

Arduino counts the number of free spaces and displays the result on the LCD.

```
SMART PARKING
```

```
Available Slots: 2
```

The LEDs also indicate the status of each parking slot. If parking space is available, the entrance gate can be opened using the servo motor. If all slots are occupied, the display can show:

```
PARKING FULL
```

6. Circuit Connection

A basic four-slot system can use four IR sensors.

Component	Arduino Connection
IR Sensor 1	Digital Pin 2
IR Sensor 2	Digital Pin 3
IR Sensor 3	Digital Pin 4
IR Sensor 4	Digital Pin 5
Servo Motor	Digital Pin 9
LCD	Digital pins / I2C
LEDs	Digital pins
VCC	5V
GND	GND

The IR sensors are placed near the parking slots. When a vehicle comes into the detection range, the sensor sends a signal to Arduino.

Note: Exact LCD connections depend on whether a normal 16×2 LCD or an I2C LCD is used.

7. Arduino Program

```
#include <Servo.h>

Servo gate;

int sensor1 = 2;
int sensor2 = 3;
int sensor3 = 4;
int sensor4 = 5;

void setup() {
  pinMode(sensor1, INPUT);
  pinMode(sensor2, INPUT);
  pinMode(sensor3, INPUT);
  pinMode(sensor4, INPUT);

  gate.attach(9);
  Serial.begin(9600);
}

void loop() {
  int occupied = 0;

  if (digitalRead(sensor1) == LOW) occupied++;
  if (digitalRead(sensor2) == LOW) occupied++;
  if (digitalRead(sensor3) == LOW) occupied++;
  if (digitalRead(sensor4) == LOW) occupied++;

  int available = 4 - occupied;

  Serial.print("Available Slots: ");
  Serial.println(available);

  if (available > 0)
    gate.write(90);
  else
    gate.write(0);

  delay(1000);
}
```

Note: The sensor logic may be HIGH/LOW depending on the IR sensor module. It can be adjusted during testing.

8. Advantages and Applications

Advantages

1. Simple design.
2. Low-cost components.
3. Easy to operate.
4. Reduces parking search time.
5. Reduces manual work.
6. Provides parking-slot information.
7. Saves fuel.
8. Can be expanded to more parking slots.

Applications

The Smart Parking System can be used in colleges, hospitals, shopping malls, offices, hotels, airports, railway stations, apartment parking areas, and small commercial parking areas.

9. Future Scope and Conclusion

Future Scope

The project can be improved in the future by adding automatic parking reservation, mobile application support, online payment, number plate recognition, more parking slots, solar power supply, and automatic parking guidance.

These features are optional. The basic project does **not** require a camera or IoT.

Conclusion

The **Smart Parking System** is a simple and useful automation project for managing parking spaces. It uses IR sensors, Arduino, LCD, LEDs, and a servo motor to detect and manage parking slots.

The system can identify occupied and available spaces and provide information to users. It reduces manual effort, saves time, and helps organize parking areas efficiently.

Therefore, the Smart Parking System is an economical and practical solution for small and medium-sized parking areas.